Charge + Light Foundation Model for MicroBooNE Open Data: Fairness Controls for the Reference Comparison

Technical Note — Round 3

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Abstract

In round 2, our multi-modal masked autoencoder qlfm-M, which takes TPC charge and PMT light together, was the best model on all four MicroBooNE tasks. A review asked whether that comparison was fair. Four concerns stood out:

- the reference models received light only through a weak late-fusion head;
- there was no charge-only qlfm at matched size;
- qlfm has an extra per-hit embedding for the hit task;
- FM4NPP was 4× smaller and capped at 16,384 hits per event.

We ran four controls:

1. a strong late-fusion head (cross-attention from charge tokens to PMT tokens), identical for every backbone;
2. a 21M charge-only qlfm (qlfm-M C) with the same pretraining as qlfm-M CL;
3. a common per-hit adapter for every backbone, plus qlfm without its hit embedding;
4. FM4NPP without the hit cap, and FM4NPP m4 (19.5M) pretrained for equal node-hours.

Findings:

- **Strong late fusion substantially improves the references,** by up to +0.21 hit F1. With it, Panda matches qlfm-M on flavor and energy at 10% labels.
- **qlfm-M keeps a clear lead at 1% labels and on the hit task** (0.90 vs 0.79 hit F1).
- **Without light in pretraining, the same 21M qlfm falls behind Panda on every task.** qlfm-M's advantage therefore comes from *fusing light during pretraining*, not from its charge architecture.
- **The hit embedding, FM4NPP's hit cap and its size explain little.**

Round 3 used 20.5 Polaris node-hours.

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*Prepared with the assistance of Claude (Anthropic). Code: `https://github.com/HaiwangYu/fm4npp-lartpc`, branch `main`, package `multi-modal-qlmatch/qlfm`. Tracking: GitHub issue #5. Round 1 (the qlfm model) and round 2 (the comparison with Pandora and four foundation-model recipes) are described in `docs/technote-round1.pdf` and `docs/technote-round2.pdf`.

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1 Summary of findings

1. **The round-2 late-light baseline understated the references.** An identical, stronger late-fusion head (Sec. 3.1) raises every reference backbone, most on the per-hit task. Frozen, 1% labels, hit F1:

   - Panda 0.715 → 0.792;
   - PoLAr-MAE 0.537 → 0.712;
   - FM4NPP 0.489 → 0.700;
   - qlfm-S C 0.429 → 0.676.

   Energy improves at 10% labels, e.g. Panda 0.302 → 0.272. Event-level ν presence changes little (all at 0.96–0.99).

2. **qlfm-M CL still leads, but the margin depends on the label budget.** Against the best fused reference, Panda + strong fusion:

	1% labels (qlfm-M vs Panda)	10% labels (qlfm-M vs Panda)
T1 hit F1	**0.899** vs 0.792	**0.914** vs 0.823
T3 flavor F1	**0.729** vs 0.706	0.764 vs 0.755 (tie)
T4 energy res. (lower better)	**0.292** vs 0.336	0.277 vs **0.272** (tie)
T2 event AUC	**0.998** vs 0.985	**0.999** vs 0.990

   (qlfm-M is quoted with strong fusion too; its own late-light numbers are within ±0.02.) So at 10% labels, Panda matches qlfm-M on flavor and energy.

3. **The decisive ingredient is light in pretraining.** qlfm-M C has the same 21M model, tokenizer, objective and 2 node-hours of pretraining as qlfm-M CL, but no light. Frozen at 1%:

	T2 AUC	T3 F1	T4 res.	T1 F1
qlfm-M CL	0.999	0.713	0.284	0.892
qlfm-M C	0.963	0.561	0.365	0.685
Panda (charge only)	0.978	0.679	0.343	0.715

   qlfm-M C is below Panda on all four tasks. Without any late light, its T2 drops to 0.723 against 0.914 for Panda.

   - **As a charge encoder, Panda is better than our cell-token Transformer.**
   - **qlfm-M's round-2 lead comes from pretraining charge and light jointly.**

4. **qlfm's per-hit embedding does not explain its hit-task lead.** qlfm-M CL scores 0.892 with its own pretrained hit embedding, 0.892 with the common raw adapter, and 0.891 with no per-hit input. A common raw adapter does help the references (Panda +0.03, PoLAr-MAE +0.07, FM4NPP +0.06), but they stay far below 0.89.

5. **FM4NPP's weak results are not caused by its hit cap or its size.**

   - Removing the 16,384-hit cap lifts frozen hit F1 from 0.463 to 0.489 (fine-tuned 1%: 0.620 → 0.654).
   - Except for a single-seed dip of m3 flavor at 10% (0.557 → 0.501), the event tasks are within ±0.02.
   - The 4× larger m4 (19.5M) at equal node-hours is at most slightly better than m3.
     - Hit F1: +0.02 frozen, +0.03 fine-tuned (0.685 vs 0.654), +0.04 with the raw adapter, +0.08 without light.

– Flavor at 10%: 0.571 vs 0.501, where m3's value is the dip above.
– Fine-tuned energy is worse (0.409 vs 0.351).

It stays far below Panda and qlfm-M CL.

2 The fairness questions

Round 2 compared qlfm with the open-data Pandora reconstruction and with four foundation-model recipes: FM4NPP, Panda, PoLAr-MAE and qlfm-SD. All got equal pretraining node-hours, identical heads, label budgets and test sets. Its main limitation is *how light enters each model* (Table 1).

Table 1: How each model uses PMT light. "LateLight" is the round-2 head module; "strong fusion" is the round-3 control (Sec. 3.1).

Model	Light in the pretrained backbone	Light in the downstream head
qlfm-S/M CL	32 PMT tokens attend jointly with the charge tokens in every layer; pretrained by masked light reconstruction (40% of events: all PMTs masked) on 747k events	LateLight (round 2) or strong fusion (round 3)
qlfm-S C, qlfm-M C	none	same
Panda, PoLAr-MAE, FM4NPP	none (their published recipes are charge-only)	same

The round-2 LateLight head is weak:

- **T2–T4:** the 32 PMT tokens are added to the token set of one attention-pooling layer, so charge and light never interact before pooling.
- **T1:** each hit receives only the event-*mean* light vector.

This favours early fusion by construction. The review raised further points:

- there was no 21M charge-only qlfm, so fusion and architecture were confounded;
- qlfm's hit head receives a pretrained 128-d per-hit embedding;
- FM4NPP was run at 4.9M parameters and capped at 16,384 hits per event. 47% of T1 test events exceed the cap, so 9.8% of the ν charge got no token and was scored as cosmic.

3 Controls

3.1 Strong late fusion (XFusion)

XFusion is identical for every backbone:

1. Token features are projected to a common 256-d width (LayerNorm + Linear).
2. Each token cross-attends to the 32 PMT tokens of its event, for 2 layers (8 heads, pre-LN, MLP ratio 2, residual).
3. The standard heads follow: AttnPool over the fused charge tokens + PMT tokens for T2–T4; for T1, the hit head uses the fused feature of the hit's token plus the mean light.

The PMT tokens come from the same LateLight embedding (per-PMT time profile, PMT position and ID) at 256-d.

- **Size:** about 1.2M trainable parameters for every backbone, 1.15M for 384-d inputs and 1.24M for Panda's 704-d.

- **Training:** from labels only, with the backbone frozen.
- **Runs:** frozen, 1% and 10% labels, all four tasks, every backbone.

3.2 qlfm-M C: charge-only qlfm at 21M

qlfm-M C is identical to qlfm-M CL (384-d, 12 layers, 6 heads, cell tokens, MAE with strip + random masking, 34k steps × 256 events, 2.0 node-hours) except `--light 0`. It is evaluated with the full R2P protocol plus the round-3 controls.

3.3 Per-hit input for T1

- **Native:** qlfm passes its pretrained per-hit embedding (in-cell offset, log charge, rms → 128-d) to the hit head. The other backbones have no such input:
  - FM4NPP has one token per hit;
  - Panda gives each hit its 0.3 cm voxel feature;
  - PoLAr-MAE gives a hit only the token of the nearest 4 cm group.
- **Raw adapter:** the same trainable MLP for every backbone, mapping plane one-hot, normalized wire, tick, log charge and rms to 64-d, replacing the native embedding.
- **None:** qlfm with no per-hit input, so hits in the same cell share a feature.
- All frozen at 1% labels, with round-2 late light.

3.4 FM4NPP: no hit cap, and m4

- **No cap downstream:** every hit gets a token. Events are micro-batched (8, or 1 when fine-tuning m4), because a forward pass over more than about 10^9 elements crashes the Mamba2 kernel.
- **m3 uncapped:** the full R2P matrix is re-run.
- **m4:** the paper's "m4" size (width 512, d_{state} 32, 12 layers, 19.5M parameters), pretrained with the m3 recipe for the same budget: 3.47 h of training, 36,700 steps, 1.17M events, 3.4 node-hours. m3 saw 2.05M events.

3.5 Protocol and compute

Everything else follows round 2 (R2P):

- same splits, label pools and test subsets (10k events; 3k for T1);
- 3,000-step cap; batch 32;
- AdamW, lr 10^{-3} frozen and 2×10^{-4} fine-tuned (layer decay 0.9);
- best-on-validation checkpoint.

Round 3 used 20.5 node-hours (`docs/figures/round3_node_hours.md`): 5.4 for pretraining (qlfm-M C 2.0, FM4NPP m4 3.4), the rest for 160 downstream runs and a smoke test.

4 Results

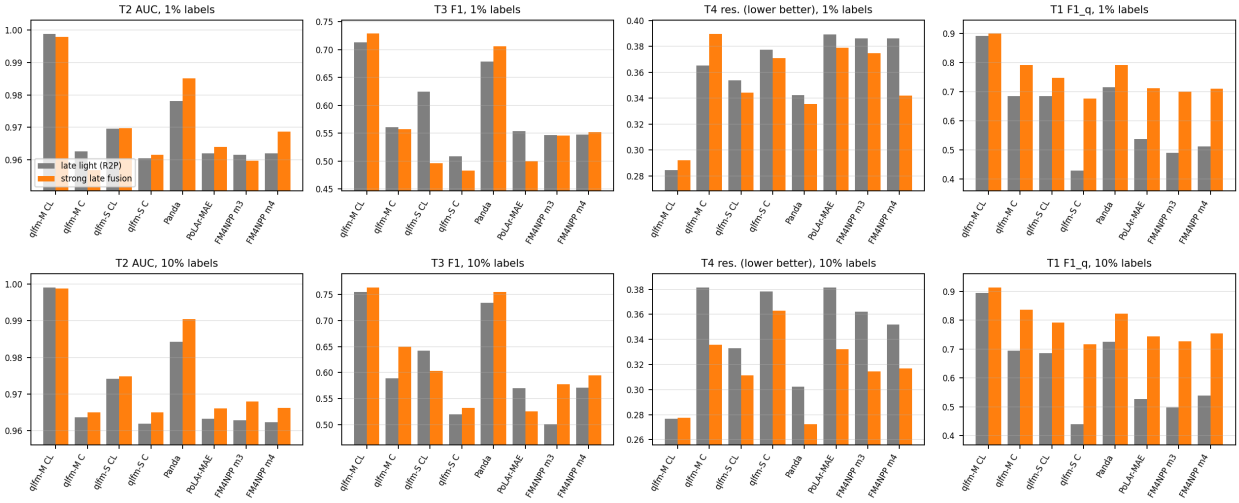


Figure 1: Frozen backbones with the round-2 late light (grey) and with strong late fusion (orange), at 1% (top) and 10% labels (bottom).

Table 2: Strong late fusion. Frozen backbone, 1% labels. LL = round-2 late light, XF = strong late fusion. Best per task in bold. "FM4NPP m3, capped" is the round-2 run; all other FM4NPP rows are uncapped.

	T2 AUC		T3 macro-F1		T4 res. (↓)		T1 $F1_q$	
Backbone	LL	XF	LL	XF	LL	XF	LL	XF
qlfm-M CL	**0.999**	0.998	0.713	**0.729**	**0.284**	0.292	0.892	**0.899**
qlfm-M C	0.963	0.957	0.561	0.558	0.365	0.390	0.685	0.792
qlfm-S CL	0.970	0.970	0.624	0.496	0.354	0.344	0.685	0.747
qlfm-S C	0.960	0.962	0.509	0.483	0.378	0.371	0.429	0.676
Panda	0.978	0.985	0.679	0.706	0.343	0.336	0.715	0.792
PoLAr-MAE	0.962	0.964	0.554	0.500	0.389	0.379	0.537	0.712
FM4NPP m3, capped	0.961	–	0.546	–	0.369	–	0.463	–
FM4NPP m3	0.961	0.960	0.546	0.546	0.386	0.375	0.489	0.700
FM4NPP m4	0.962	0.969	0.548	0.552	0.386	0.342	0.511	0.710

Table 3: As Table 2, at 10% labels.

	T2 AUC		T3 macro-F1		T4 res. (↓)		T1 $F1_q$	
Backbone	LL	XF	LL	XF	LL	XF	LL	XF
qlfm-M CL	**0.999**	**0.999**	0.755	**0.764**	0.276	0.277	0.894	**0.914**
qlfm-M C	0.964	0.965	0.589	0.650	0.382	0.336	0.694	0.837
qlfm-S CL	0.974	0.975	0.642	0.603	0.333	0.311	0.685	0.792
qlfm-S C	0.962	0.965	0.520	0.533	0.378	0.363	0.440	0.717
Panda	0.984	0.990	0.734	0.755	0.302	**0.272**	0.725	0.823
PoLAr-MAE	0.963	0.966	0.570	0.526	0.381	0.332	0.526	0.744
FM4NPP m3, capped	0.963	–	0.557	–	0.358	–	0.468	–
FM4NPP m3	0.963	0.968	0.501	0.578	0.362	0.315	0.497	0.727
FM4NPP m4	0.962	0.966	0.571	0.595	0.352	0.317	0.539	0.754

Table 4: Full R2P matrix for the new backbones (qlfm-M C, FM4NPP m3 uncapped, FM4NPP m4) next to the round-2 rows. "no light" is the frozen run without late light; for qlfm-M CL the light inside the backbone remains. ±: spread over 2 seeds.

T2 event ν/cosmic, AUC

Backbone	frozen 0.1%	frozen 1%	frozen 10%	FT 0.1%	FT 1%	no light, 1%
qlfm-M CL	**0.986±.002**	**0.999**	**0.999**	**0.985**	**0.998**	**0.999**
qlfm-M C	0.945±.002	0.963	0.964	0.656	0.960	0.723
qlfm-S C	0.943±.001	0.960	0.962	0.664	0.956	0.666
Panda	0.938±.007	0.978	0.984	0.830	0.968	0.914
FM4NPP m3, capped	0.943±.000	0.961	0.963	0.626	0.944	0.712
FM4NPP m3	0.944±.001	0.961	0.963	0.624	0.947	0.719
FM4NPP m4	0.943±.002	0.962	0.962	0.650	0.965	0.743

T3 flavor, macro-F1

Backbone	frozen 0.1%	frozen 1%	frozen 10%	FT 0.1%	FT 1%	no light, 1%
qlfm-M CL	**0.531±.003**	**0.713**	**0.755**	**0.640**	**0.724**	**0.709**
qlfm-M C	0.443±.001	0.561	0.589	0.412	0.507	0.510
qlfm-S C	0.457±.000	0.509	0.520	0.389	0.469	0.429
Panda	0.465±.001	0.679	0.734	0.506	0.690	0.666
FM4NPP m3, capped	0.445±.003	0.546	0.557	0.355	0.485	0.507
FM4NPP m3	0.454±.005	0.546	0.501	0.358	0.509	0.498
FM4NPP m4	0.427±.004	0.548	0.571	0.342	0.534	0.530

Table 5: Full R2P matrix (continued).

T4 energy, resolution (lower is better)

Backbone	frozen 0.1%	frozen 1%	frozen 10%	FT 0.1%	FT 1%	no light, 1%
qlfm-M CL	**0.418±.003**	**0.284**	**0.276**	**0.385**	**0.303**	**0.284**
qlfm-M C	0.465±.001	0.365	0.382	0.563	0.412	0.419
qlfm-S C	0.484±.001	0.378	0.378	0.563	0.412	0.441
Panda	0.496±.001	0.343	0.302	0.484	0.358	0.423
FM4NPP m3, capped	0.454±.013	0.369	0.358	0.567	0.334	0.420
FM4NPP m3	0.456±.010	0.386	0.362	0.565	0.351	0.410
FM4NPP m4	0.473±.008	0.386	0.352	0.522	0.409	0.395

T1 hit ν/cosmic, F1$_q$

Backbone	frozen 0.1%	frozen 1%	frozen 10%	FT 0.1%	FT 1%	no light, 1%
qlfm-M CL	**0.868±.000**	**0.892**	**0.894**	**0.905**	**0.947**	**0.891**
qlfm-M C	0.334±.020	0.685	0.694	0.532	0.889	0.561
qlfm-S C	0.080±.001	0.429	0.440	0.008	0.828	0.170
Panda	0.536±.003	0.715	0.725	0.656	0.823	0.629
FM4NPP m3, capped	0.200±.013	0.463	0.468	0.371	0.620	0.238
FM4NPP m3	0.170±.009	0.489	0.497	0.413	0.654	0.286
FM4NPP m4	0.233±.002	0.511	0.539	0.439	0.685	0.364

Table 6: T1 per-hit input, frozen at 1% labels, $F1_q$. Native: the round-2 run, i.e. qlfm's pretrained hit embedding, or no per-hit input (*) for the other backbones. Raw: common trainable adapter. None: qlfm without a per-hit input. The XF column (strong fusion, native hit input) is shown for reference.

Backbone	native	raw adapter	none	XF (native)
qlfm-M CL	0.892	0.892	0.891	0.899
qlfm-M C	0.685	0.706	0.682	0.792
qlfm-S CL	0.685	0.686	0.685	0.747
qlfm-S C	0.429	0.482	0.395	0.676
Panda	0.715*	0.741	–	0.792
PoLAr-MAE	0.537*	0.603	–	0.712
FM4NPP m3	0.489*	0.546	–	0.700
FM4NPP m4	0.511*	0.581	–	0.710

5 Discussion

Was round 2 fair? In protocol yes (same data, labels, heads, budgets), in light handling no: the references could use light only through a head that cannot match charge with light.

- Once that head is strengthened, the references gain a lot, and at 10% labels Panda is on par with qlfm-M on flavor and energy.
- The round-2 statement "qlfm-M is best on all four tasks from 1% labels" still holds for 1% labels and for the hit task, but not uniformly at 10%.

Why early fusion wins where it does. ν/cosmic separation is largely a charge–light *matching* problem: which charge is in time and in position with the beam flash.

- With light inside the backbone, MAE pretraining forces the charge tokens to predict the PMT pattern. This learns the matching from 747k unlabelled events.
- A head must instead learn it from 300–7,000 labelled events. Strong late fusion closes much of the gap when labels are plentiful (10%), but not with 1% labels or for per-hit decisions.
- This also explains why qlfm-M CL itself barely changes with strong fusion: its tokens already carry the matched light.

Which charge encoder is better. At matched size and compute, Panda (PTv3 + Sonata self-distillation) is a better charge-only encoder than qlfm's cell-token MAE. This holds with the R2P head (better on all four tasks), with strong late fusion (T2–T4 clearly better; T1 equal at 1% and 0.823 vs 0.837 at 10%), and without any light (T2 0.914 vs 0.723). FM4NPP m3, FM4NPP m4 and qlfm-M C form a second group at about the same level. A natural next step is to combine the two strengths: Panda-style charge encoding with light fused during pretraining.

Smaller effects.

- **Hit embedding:** qlfm's hit embedding is irrelevant at 21M (0.892 / 0.892 / 0.891) and matters only for the small charge-only model. A per-hit raw adapter is a cheap, general improvement for the point-cloud references (PoLAr-MAE +0.07, FM4NPP m3/m4 +0.06/+0.07).
- **FM4NPP cap:** removing it is worth ≈0.03 hit F1.
- **FM4NPP size:** at equal node-hours, 4× more parameters (m4, 1.17M events) gives at most small gains over m3 (2.05M events).
- **Strong fusion at 1% labels on flavor:** it sometimes overfits, e.g. qlfm-S CL 0.624 → 0.496 and PoLAr-MAE 0.554 → 0.500. The extra 1.2M head parameters are large for about 3k labelled events.

6 Limitations

- Single seed except frozen 0.1%; differences below about 0.02 are not significant.
- Strong fusion was run frozen only. Fine-tuning with it, and combining it with the raw hit adapter, are not tested.
- The references remain charge-only in pretraining. Giving them light during pretraining (the fair "early fusion for everyone" test) needs changes to each recipe and is left open.
- Home advantage remains: qlfm's design was iterated on this dataset in round 1; the references use one untuned adaptation each.
- The FM4NPP m4 fine-tune runs first ran out of memory at micro-batch 4. They were re-run at micro-batch 1 (same batch of 32 events); the 1% runs needed one save/resume across jobs and all reached their full step count.

7 Reproducibility

- Code (commit on `main`):
  - `qlfm/models/heads.py`: `XFusion`, `RawHit`;
  - `qlfm/train_downstream.py`: `--late-light 2`, `--hit-feat`, `--max-hits`, `--resume`;
  - `qlfm/ref/fm4npp.py`: cap override.
- Tasks: `scripts/make_tasks_r3.py OUT {xf,hit,r2p} BACKBONE...`, run by `scripts/run_queue.sh`.
- Pretraining:
  - qlfm-M C: `qlfm.train_pretrain --light 0 --d 384 --depth 12 --heads 6 --steps 34000 --bs 64`.
  - FM4NPP m4: `--method fm4npp --d 512 --d-state 32 --budget-h 3.47` (mamba venv).
- Tables and figure: `scripts/summarize_r3.py R2P_ROOT R3_ROOT docs/technote-round3-tex/gen docs/figures/round3_fusion.png`. Unfinished runs print as pending, so the note is rebuilt by re-running it.